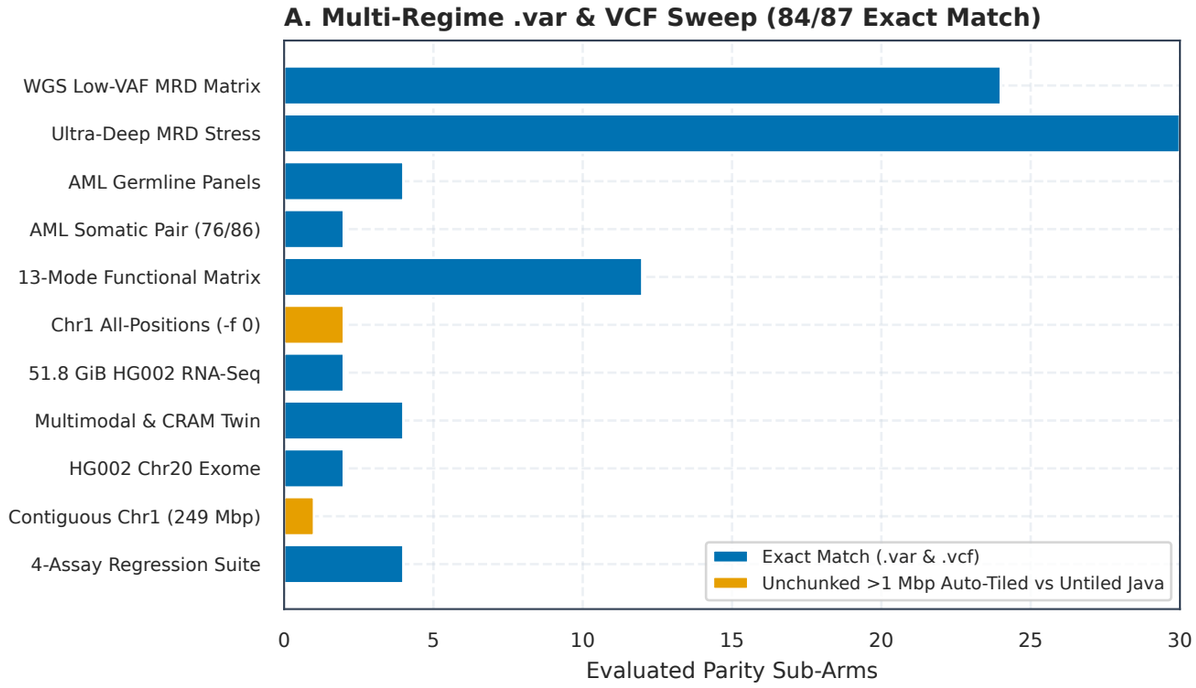
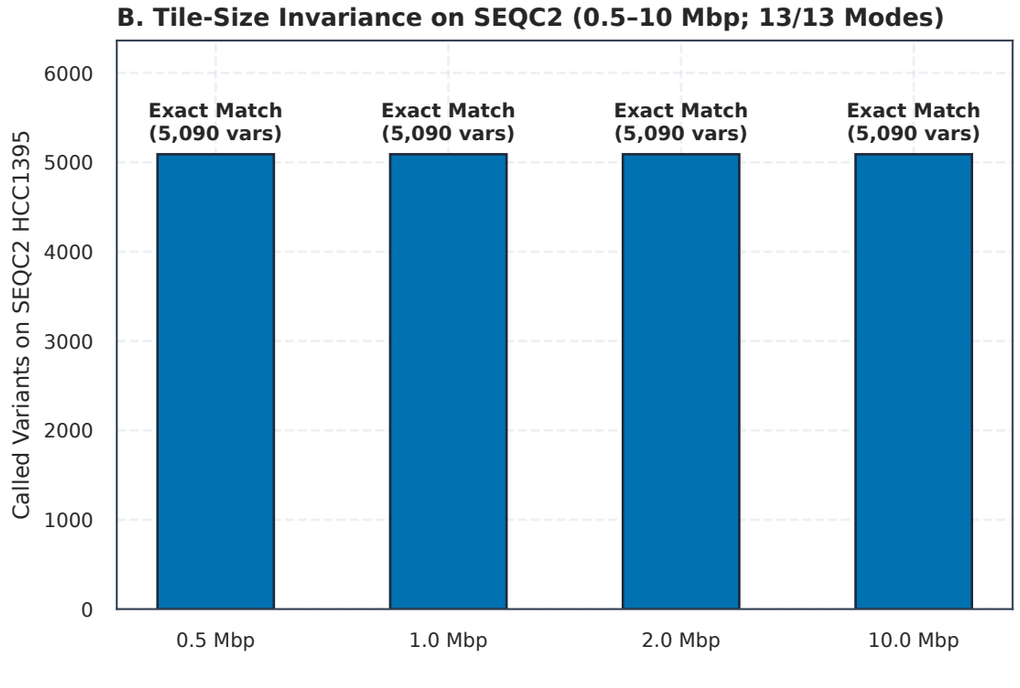
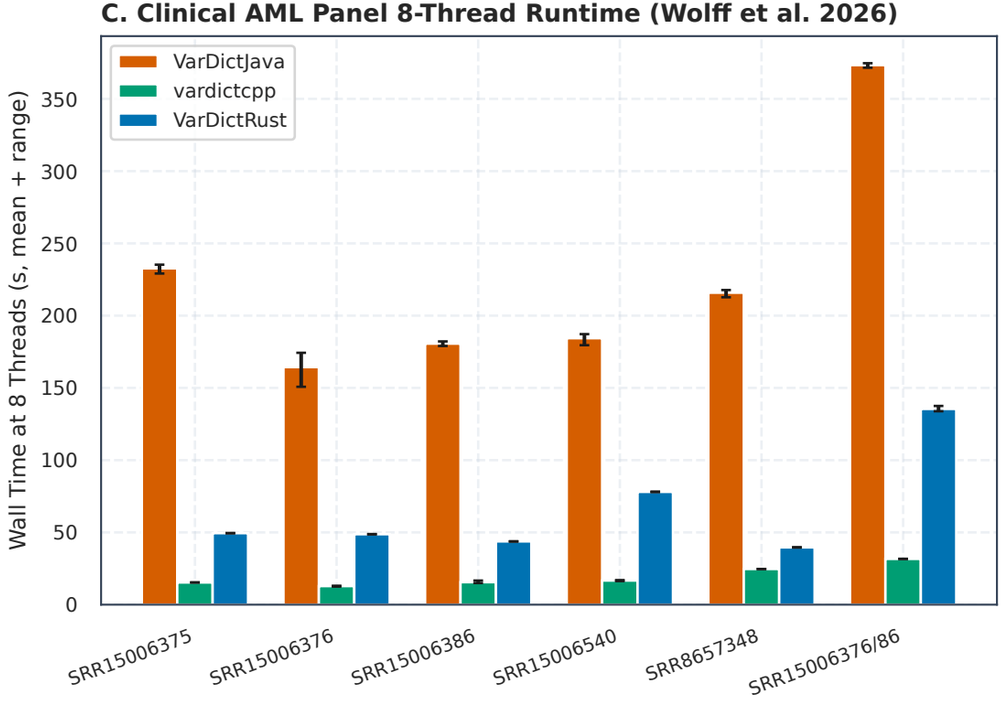
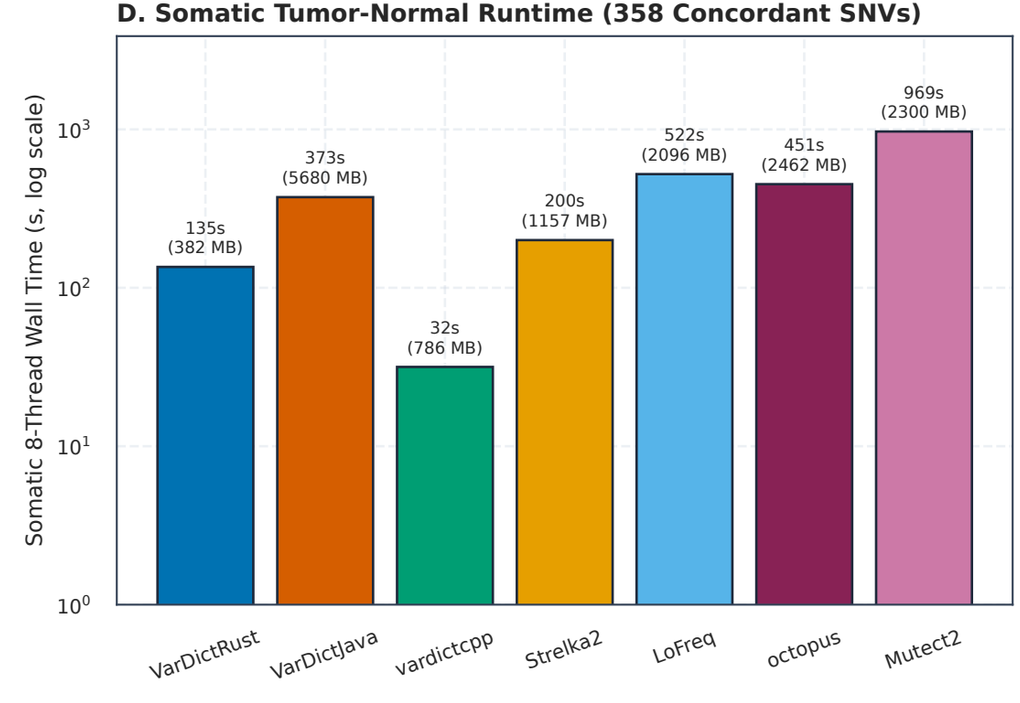

A. Multi-Regime .var & VCF Sweep (84/87 Exact Match)
WGS Low-VAF MRD Matrix
Ultra-Deep MRD Stress
AML Germline Panels
AML Somatic Pair (76/86)
13-Mode Functional Matrix
Chr1 All-Positions (-f 0)
51.8 GiB HG002 RNA-Seq
Multimodal & CRAM Twin
HG002 Chr20 Exome
Contiguous Chr1 (249 Mbp)
4-Assay Regression Suite
Exact Match (.var & .vcf)
Unchunked >1 Mbp Auto-Tiled vs Untiled Java
0
5
10
15
20
25
30
Evaluated Parity Sub-Arms


B. Tile-Size Invariance on SEQC2 (0.5–10 Mbp; 13/13 Modes)
Called Variants on SEQC2 HCC1395
6000
5000
4000
3000
2000
1000
0
Exact Match (5,090 vars)
Exact Match (5,090 vars)
Exact Match (5,090 vars)
Exact Match (5,090 vars)
0.5 Mbp
1.0 Mbp
2.0 Mbp
10.0 Mbp


C. Clinical AML Panel 8-Thread Runtime (Wolff et al. 2026)
VarDictJava
vardictcpp
VarDictRust
Wall Time at 8 Threads (s, mean + range)
350
300
250
200
150
100
50
0
SRR15006375
SRR15006376
SRR15006386
SRR15006540
SRR8657348
SRR15006376/86


D. Somatic Tumor-Normal Runtime (358 Concordant SNVs)
Somatic 8-Thread Wall Time (s, log scale)
10^3
10^2
10^1
10^0
135s (382 MB)
373s (5680 MB)
32s (786 MB)
200s (1157 MB)
522s (2096 MB)
451s (2462 MB)
969s (2300 MB)
VarDictRust
VarDictJava
vardictcpp
Strelka2
LoFreq
octopus
Mutect2